

Lunar Lander Pose Estimation

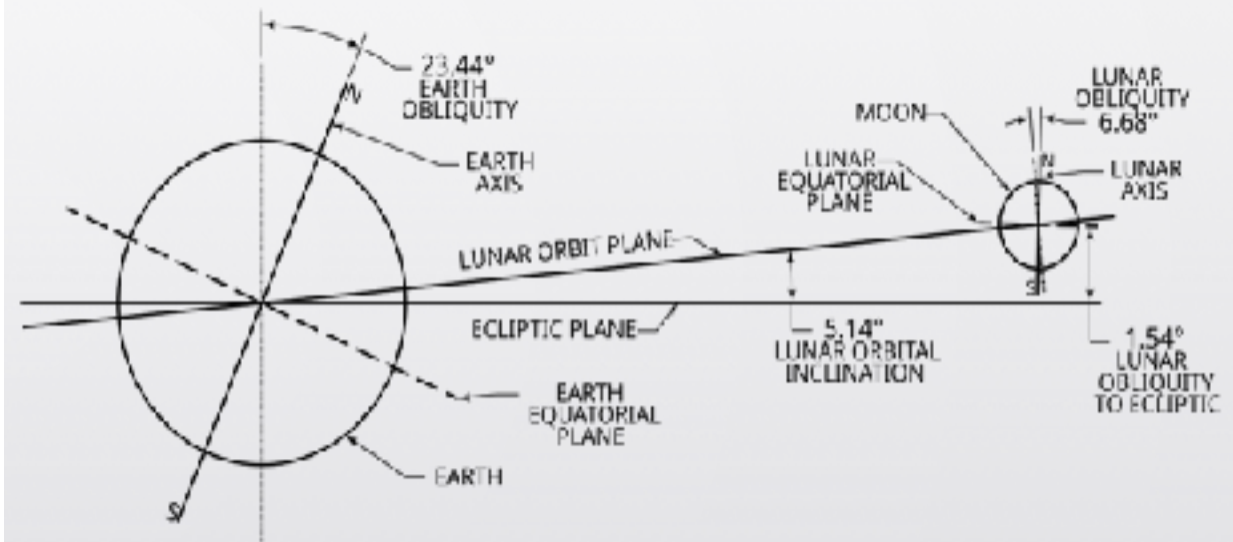
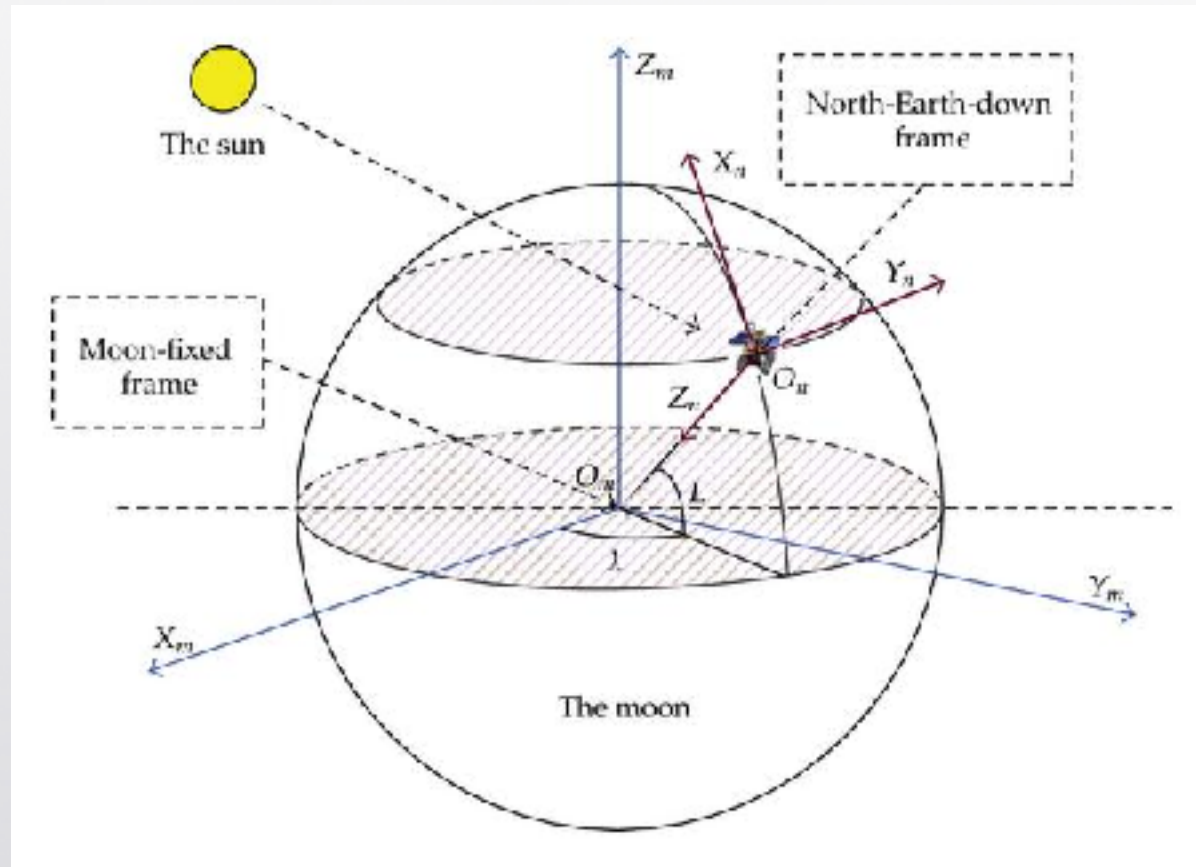
Luke Serrano



Problem Statement



Moon-Centered Moon-Fixed Coordinate System



NOTE: EARTH AND MOON RELATIVE SIZES AND ANGLES ARE TO SCALE.
EARTH AND MOON RELATIVE DISTANCE IS NOT TO SCALE.

States and Dynamics

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{r}_{B/M}^M(t) \\ \mathbf{v}_{B/M}^M(t) \\ \hat{\mathbf{q}}_M^B(t) \end{bmatrix}$$

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \mathbf{v}_{B/M}^M(t) \\ -\mu_{\text{Moon}} \frac{\mathbf{r}_{B/M}^M(t)}{\|\mathbf{r}_{B/M}^M(t)\|^3} - 2\boldsymbol{\omega}_{M/N}^M \times \mathbf{v}_{B/M}^M(t) - \boldsymbol{\omega}_{M/N}^M \times (\boldsymbol{\omega}_{M/N}^M \times \mathbf{r}_{B/M}^M(t)) \\ \frac{1}{2} \boldsymbol{\Omega}[\boldsymbol{\omega}_{B/M}^B(t)] \hat{\mathbf{q}}_M^B(t) \end{bmatrix} + \mathbf{w}(t)$$

Landmark Measurements (1 Hz)

Lunar Landmark Identification + a priori map

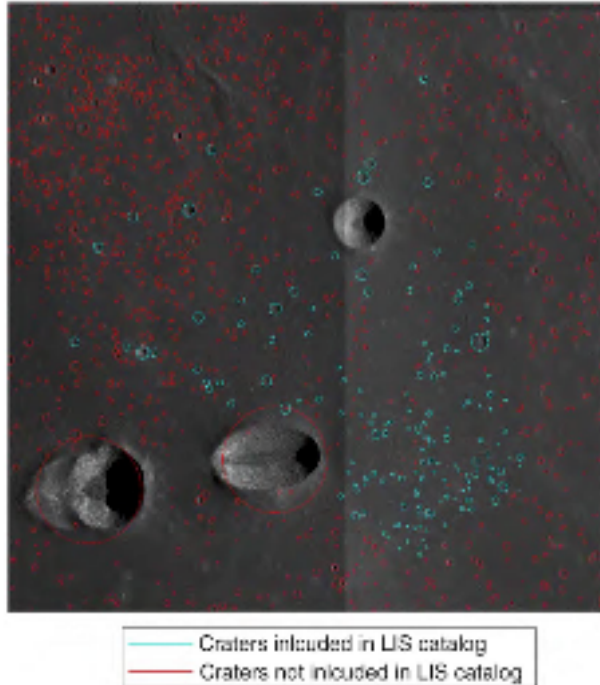
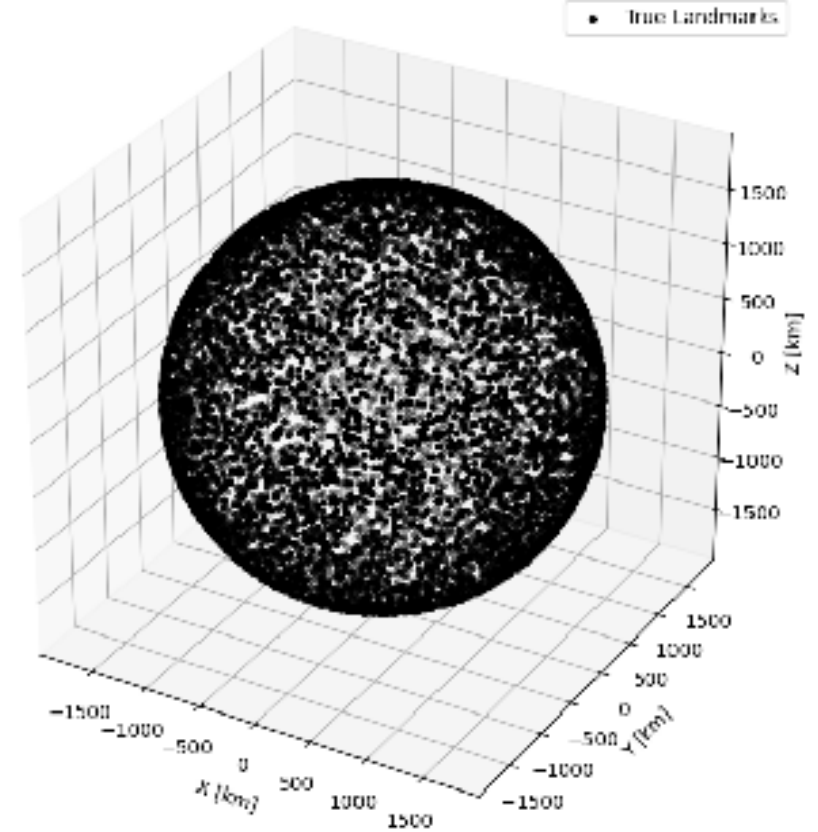


Fig. 7 Local LIS catalog selected from regional catalog based on single reference camera pose.

Maass, B., Woicke, S., Oliveira, W. M., Razgus, B., & Krüger, H. (2020). Crater navigation system for autonomous precision landing on the moon. *Journal of Guidance, Control, and Dynamics*, 43(8), 1414–1431. American Institute of Aeronautics and Astronautics.



$$\mathbf{r}_{l/C,k}^C = \tilde{\mathbf{z}}_k = \mathbf{T}_B^C \mathbf{T}_{M,k}^B (\mathbf{r}_{l/M}^M - \mathbf{r}_{B/M,k}^M) + \mathbf{v}_k$$

Translational Extended Kalman Filter (EKF)



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$$\frac{d}{dt} \begin{bmatrix} \mathbf{r}_{B/M}^M(t) \\ \mathbf{v}_{B/M}^M(t) \end{bmatrix} = \begin{bmatrix} \mathbf{v}_{B/M}^M(t) \\ -\mu_{\text{Moon}} \frac{\mathbf{r}_{B/M}^M(t)}{\|\mathbf{r}_{B/M}^M(t)\|^3} - 2\boldsymbol{\omega}_{M/N}^M \times \mathbf{v}_{B/M}^M(t) - \boldsymbol{\omega}_{M/N}^M \times (\boldsymbol{\omega}_{M/N}^M \times \mathbf{r}_{B/M}^M(t)) \end{bmatrix} + \mathbf{w}(t)$$

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{r}_{B/M}^M(t) \\ \mathbf{v}_{B/M}^M(t) \\ \alpha(t) \\ \delta\beta(t) \end{bmatrix}$$

$$\mathbf{F}_x(t) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{I}_{3 \times 3} \\ -\mu_{\text{Moon}} \left(\frac{\mathbf{I}_3}{\|\mathbf{r}_{B/M}^M\|^3} - 3 \frac{\mathbf{r}_{B/M}^M \mathbf{r}_{B/M}^{M\top}}{\|\mathbf{r}_{B/M}^M\|^5} \right) - [\boldsymbol{\omega}_{M/N}^M \times] [\boldsymbol{\omega}_{M/N}^M \times] & -2 [\boldsymbol{\omega}_{M/N}^M \times] \end{bmatrix}$$

Multiplicative Extended Kalman Filter (MEKF)



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$$\hat{\mathbf{q}}_M^B = \delta \hat{\mathbf{q}}_{B'}^B \otimes \hat{\mathbf{q}}_M^{B'}$$

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \frac{d}{dt}(\alpha) \\ \frac{d}{dt}(\delta\beta) \end{bmatrix} = \begin{bmatrix} -[\mathbf{m}_{\omega,k} \times] & -I_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} -I_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix} \begin{bmatrix} \mathbf{w}_{\omega} \\ \mathbf{w}_{\beta} \end{bmatrix}$$

$${}^* \hat{\mathbf{q}}_{Mk}^{B+} = \begin{bmatrix} \mathbf{m}_{\alpha_k^+}/2 \\ 1 \end{bmatrix} \otimes {}^* \hat{\mathbf{q}}_{Mk}^{B-} \rightarrow {}^* \hat{\mathbf{q}}_{Mk}^{B+} = \| {}^* \hat{\mathbf{q}}_{Mk}^{B+} \|, \mathbf{m}_{\alpha_{k-1}^+} = \mathbf{0}$$

$$\mathbf{m}_{\beta_k^+} = \mathbf{m}_{\delta\beta_k^+} + \mathbf{m}_{\beta_k^-}, \mathbf{m}_{\delta\beta_{k-1}^+} = \mathbf{0}$$

Extended Gaussian Mixture Filter (EGMF)



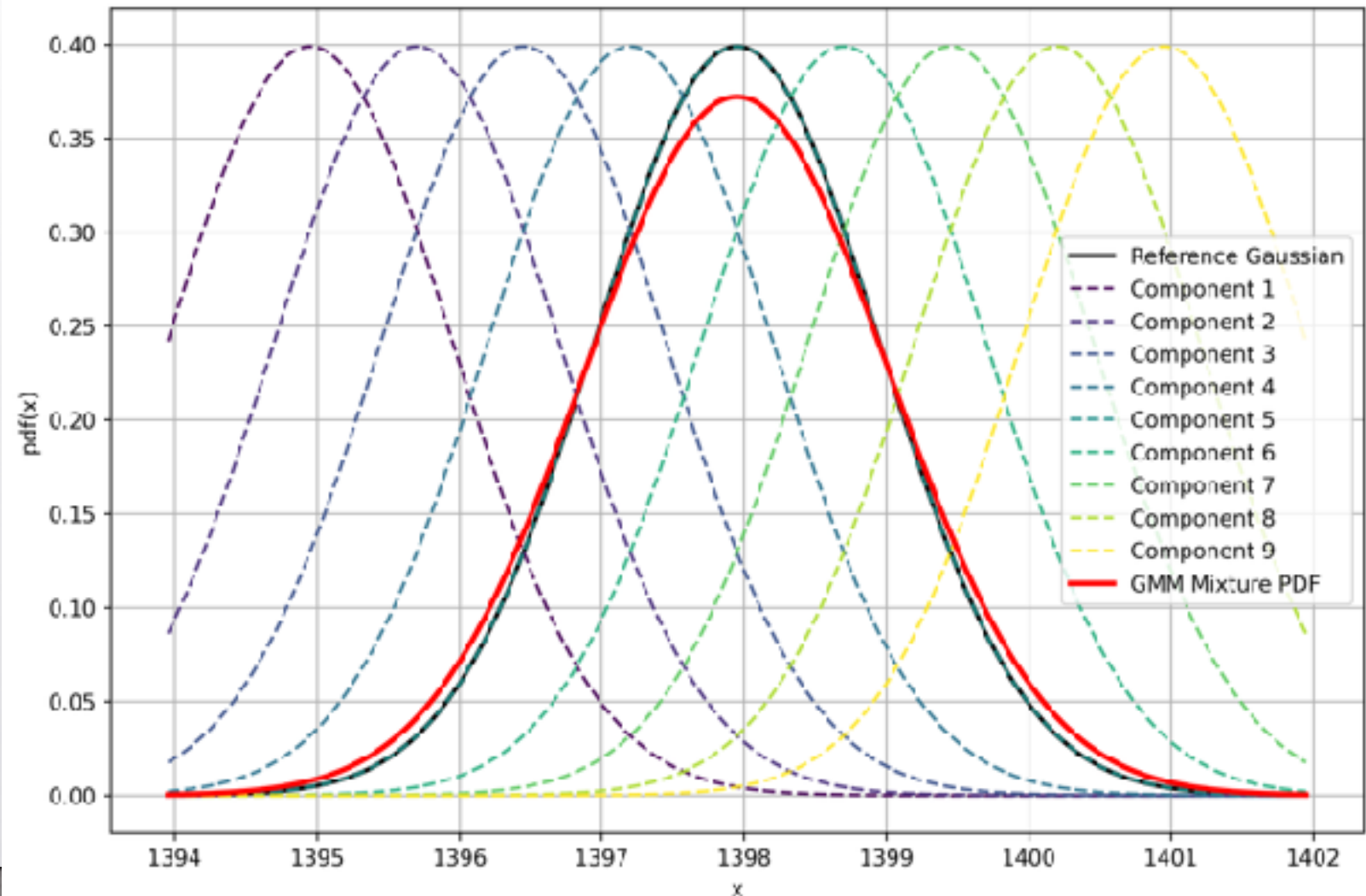
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$$\mathbf{x}_k = f(\mathbf{x}_{k-1}) + \mathbf{w}_{k-1},$$

Weight Initialization:

$$p_g(\mathbf{m}_{x,l}; \mathbf{m}_{x,\text{ref}}, \mathbf{P}_{xx,\text{ref}})$$



Simulation Environment



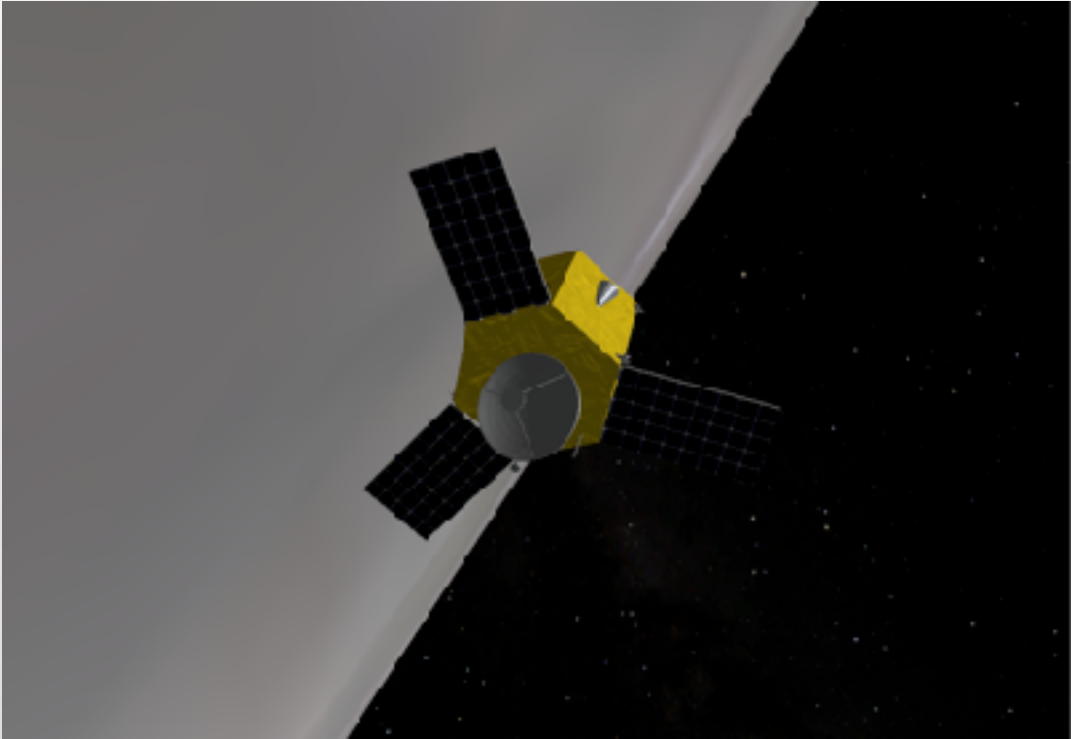
true gyroscope bias for is taken to be a constant 0.1 deg/hr

Orbital Element	Value
Semi-major axis a	$r_{\text{Lunar Equatorial Radius}} + 2,000 \text{ m}$
Eccentricity e	0.00001
Inclination i	30°
RAAN Ω	30°
Argument of Perigee ω	0°
True Anomaly ν	0°

Table 2 Initial Orbital Elements

$$\mathbf{I} = \begin{bmatrix} 900.0 & 0.0 & 0.0 \\ 0.0 & 800.0 & 0.0 \\ 0.0 & 0.0 & 600.0 \end{bmatrix} \text{ kg} \cdot \text{m}^2, \quad m = 750.0 \text{ kg}$$

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m = -G \frac{M_e m_{sc}}{r_{sc/e}^3} \mathbf{r}_{sc/e} - G \frac{M_m m_{sc}}{r_{sc/m}^3} \mathbf{r}_{sc/m}$$



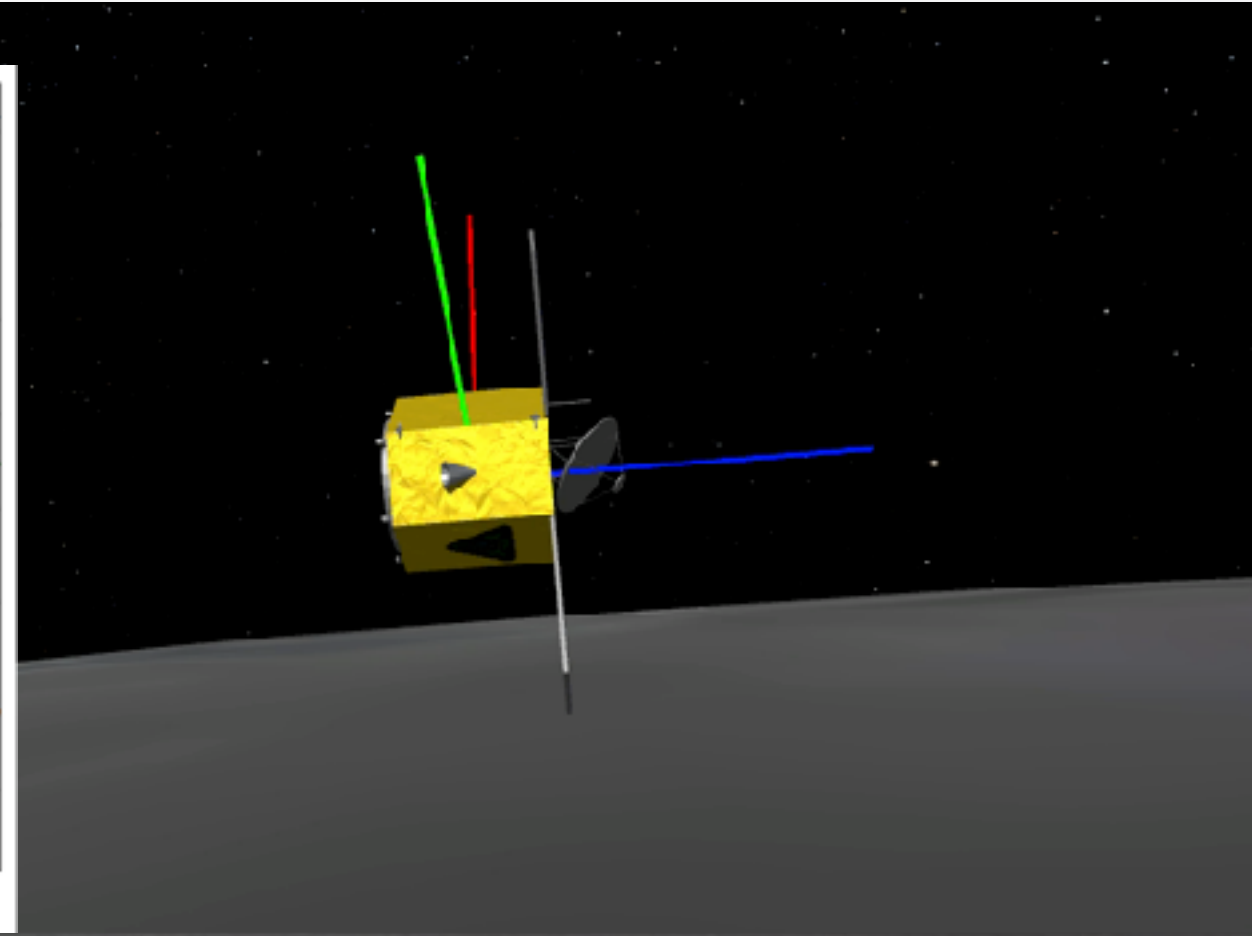
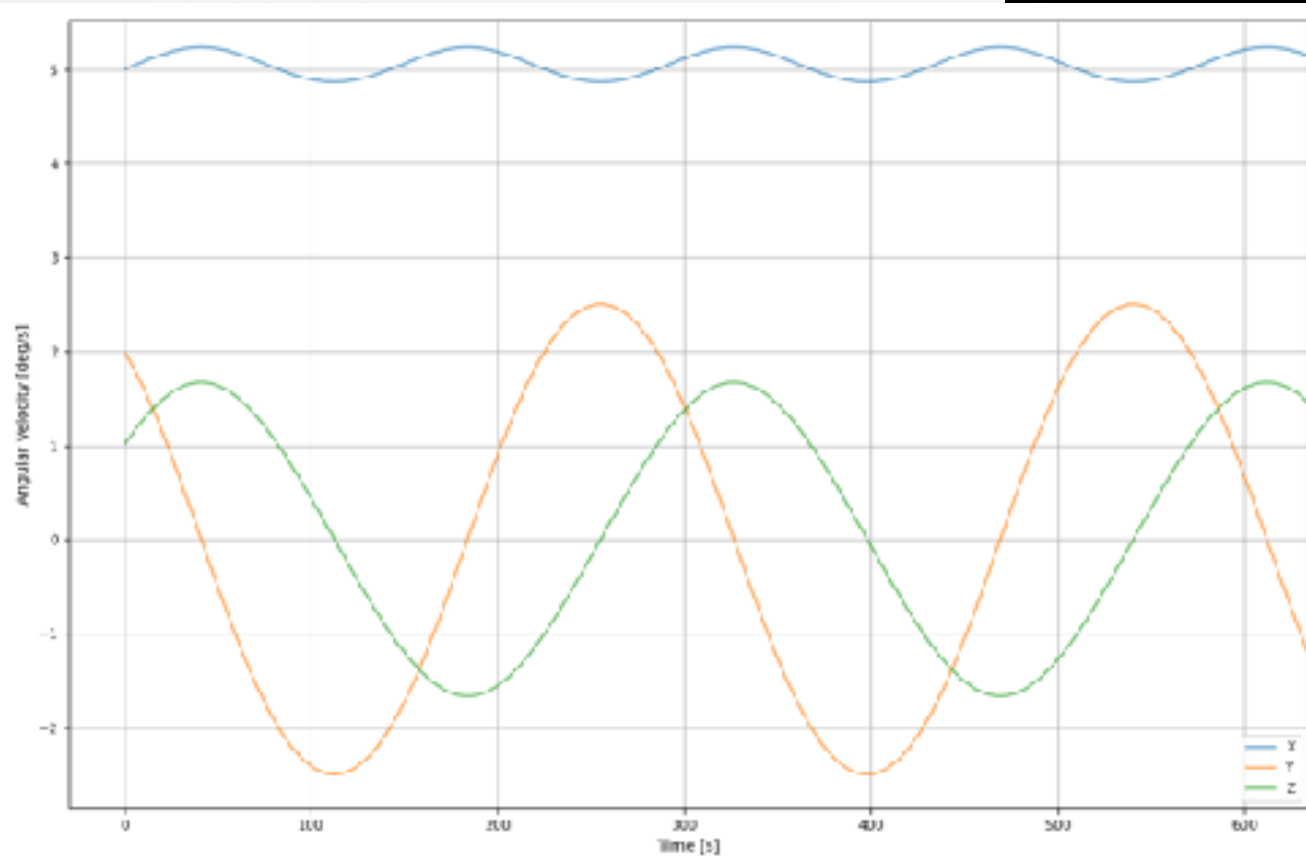
Scenario I: Passive Tumble



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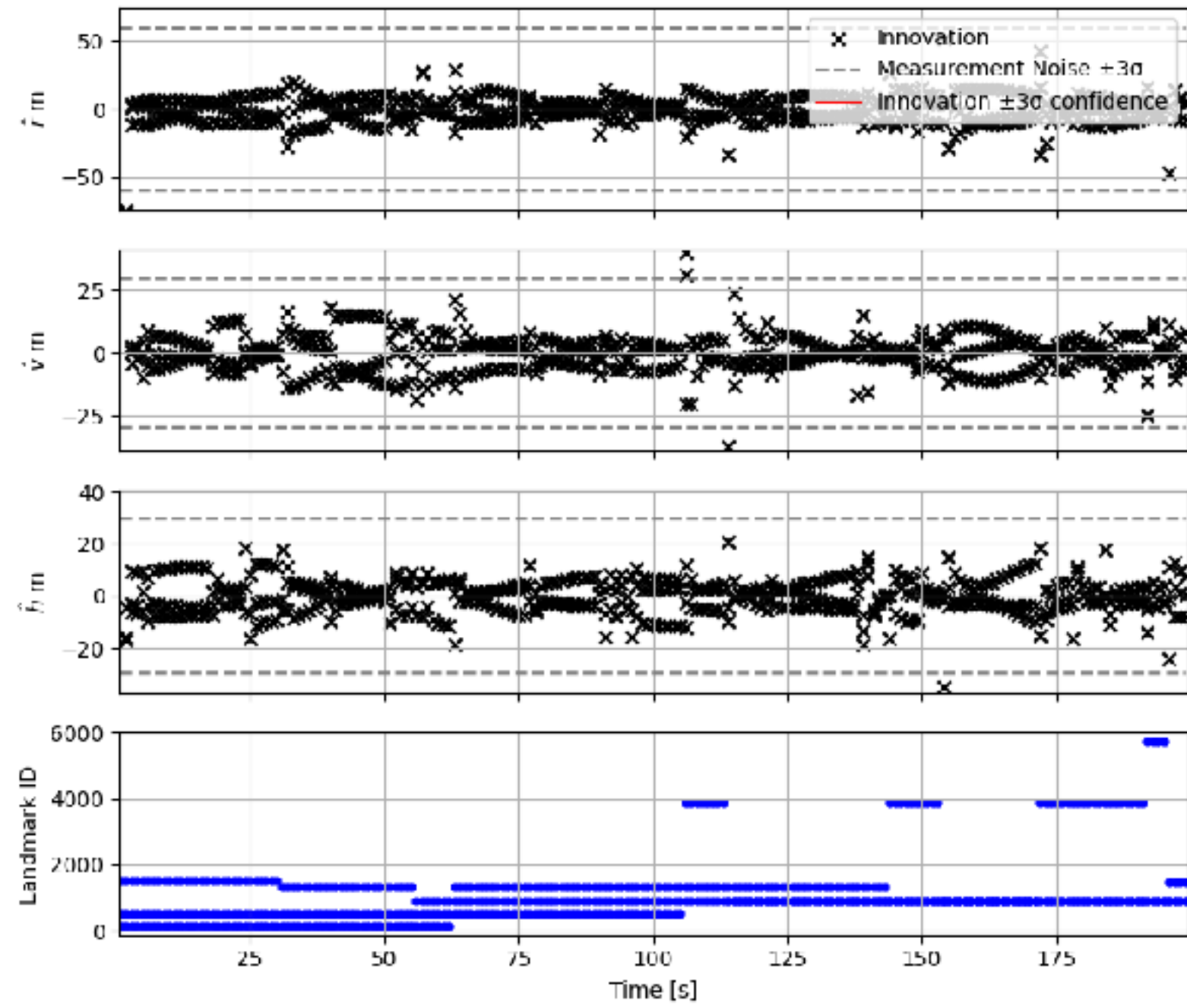
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$$\omega_{B/N}^B = [5.0, 2.0, 1.0]^T \text{ deg/s}$$

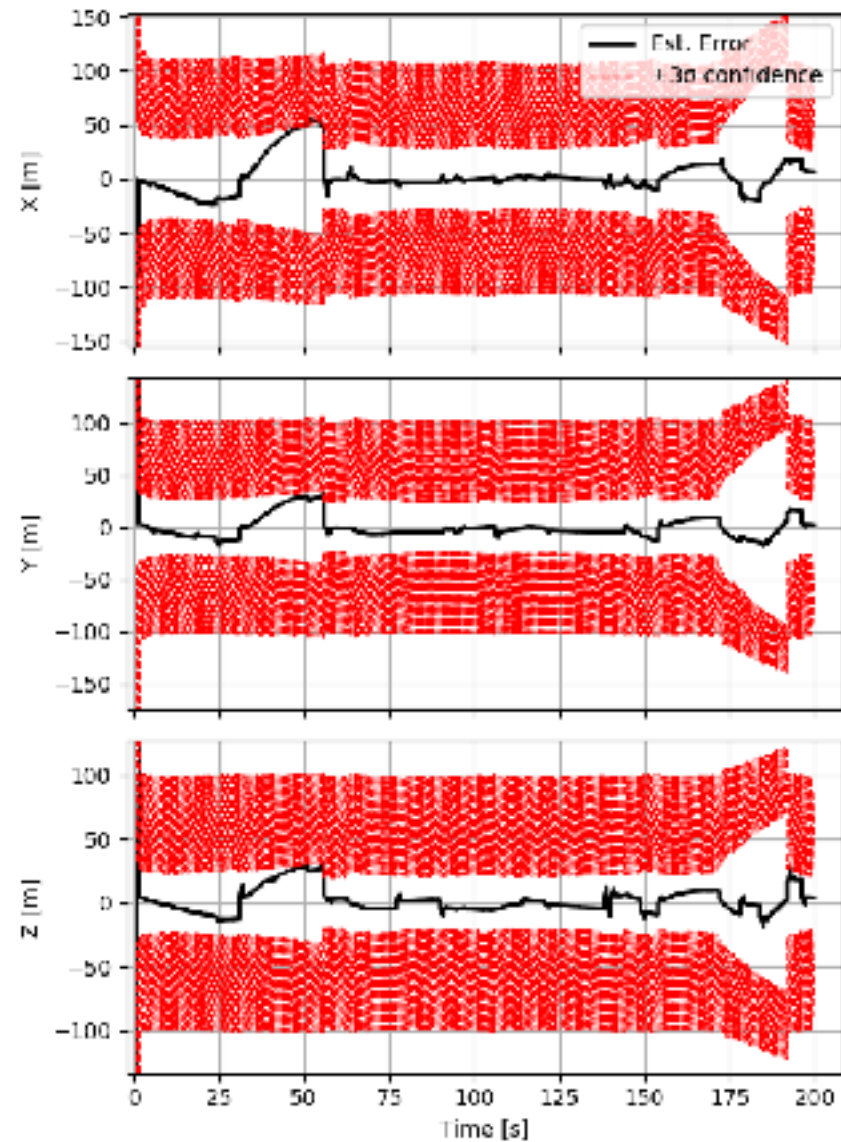


Scenario 1: EKF LVLH Frame Landmark Innovations and Landmark ID Trajectory

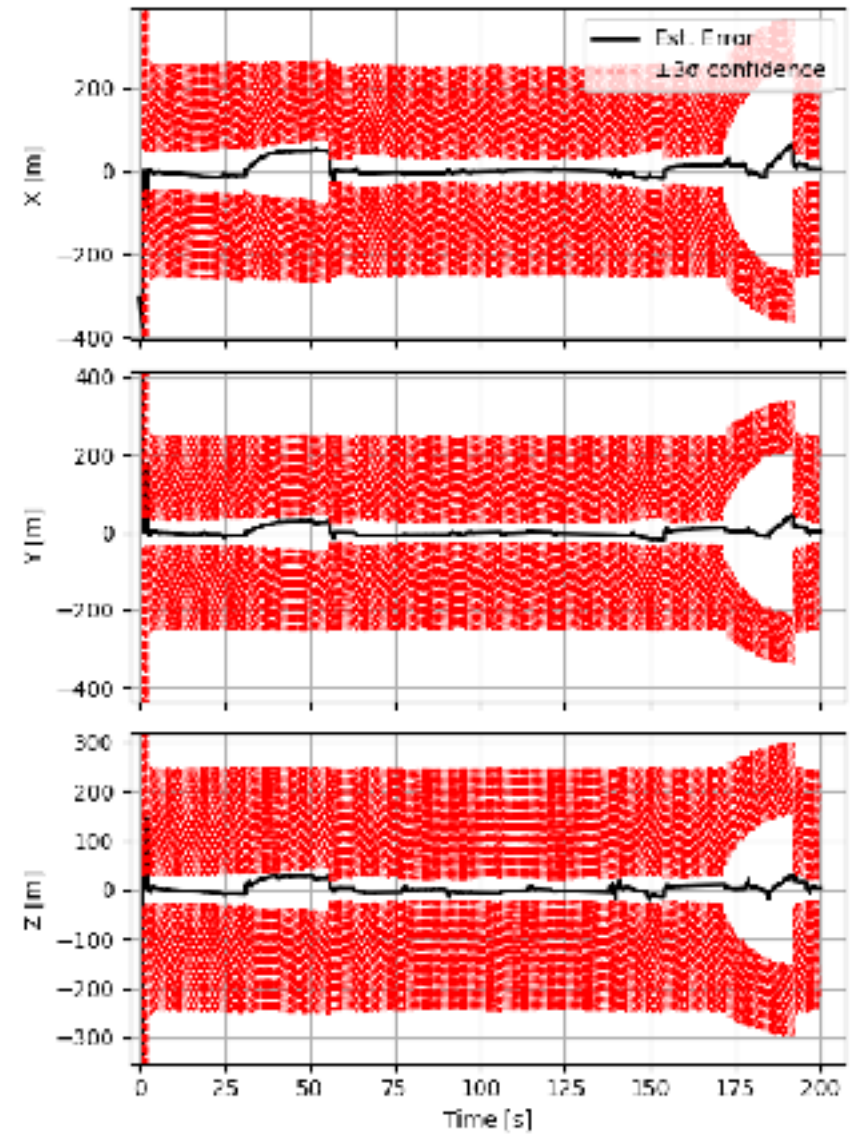
Space Robotics Laboratory



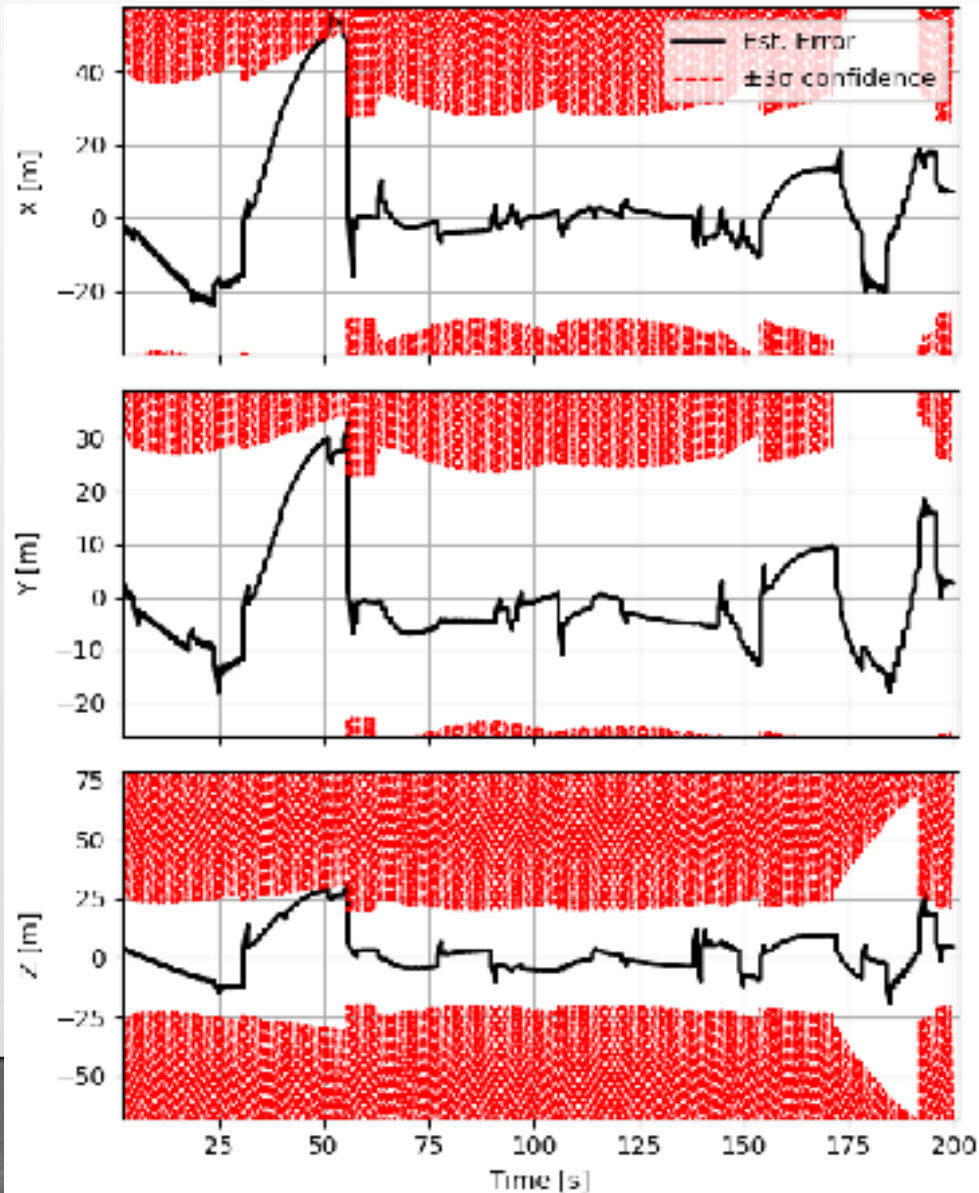
EKF



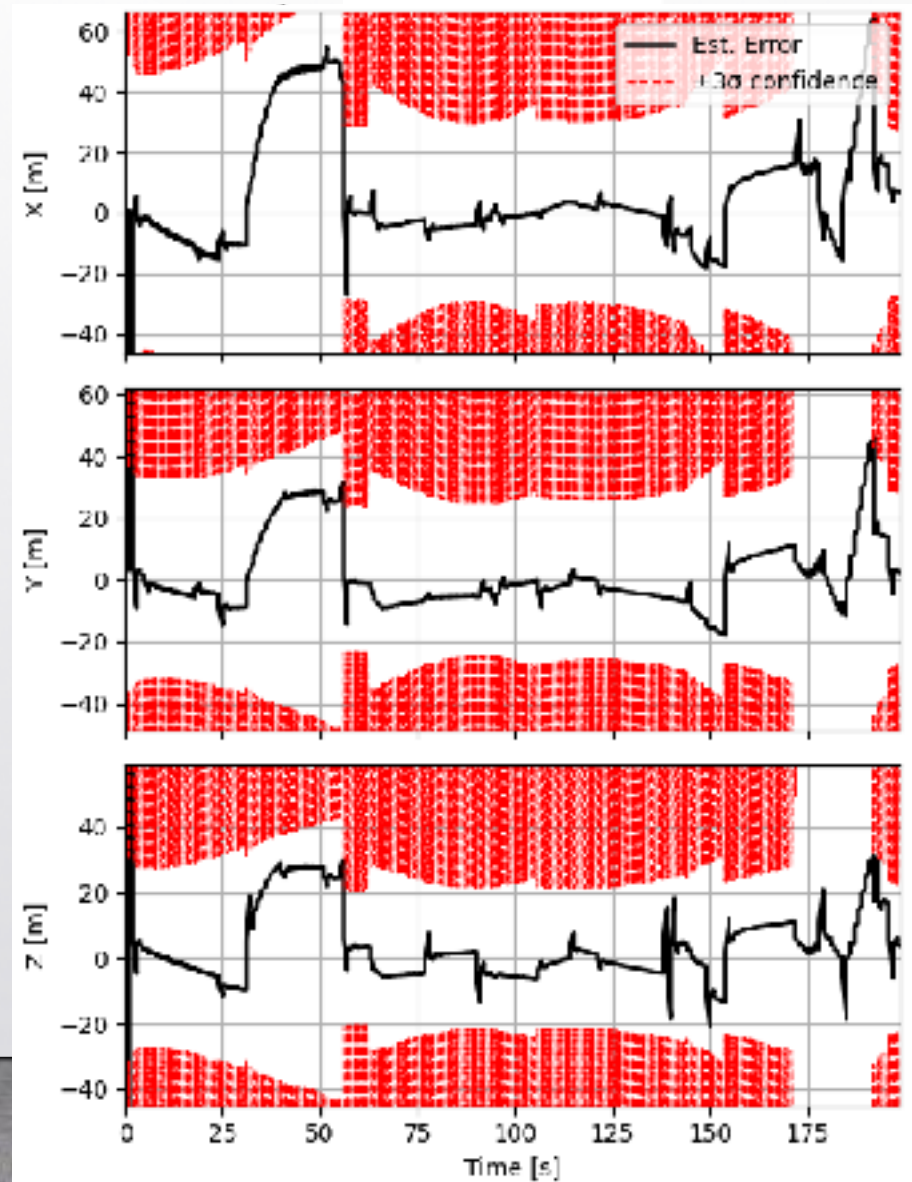
EGMF



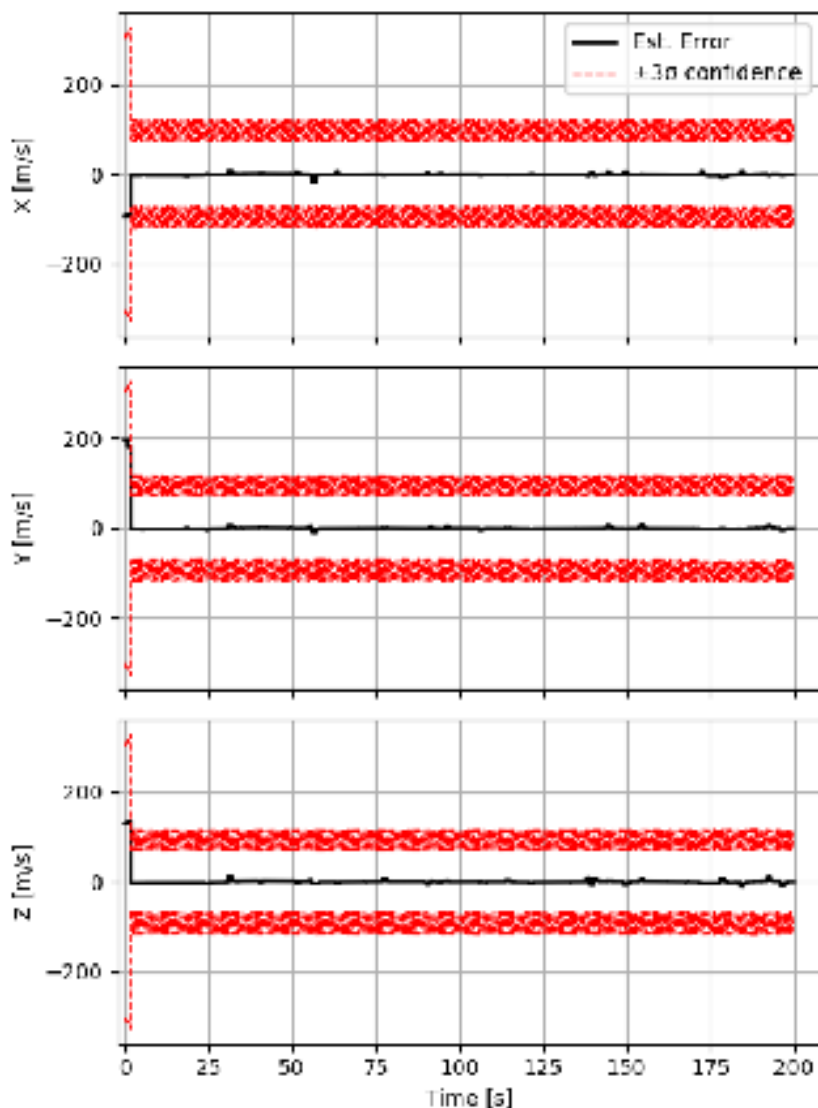
EKF



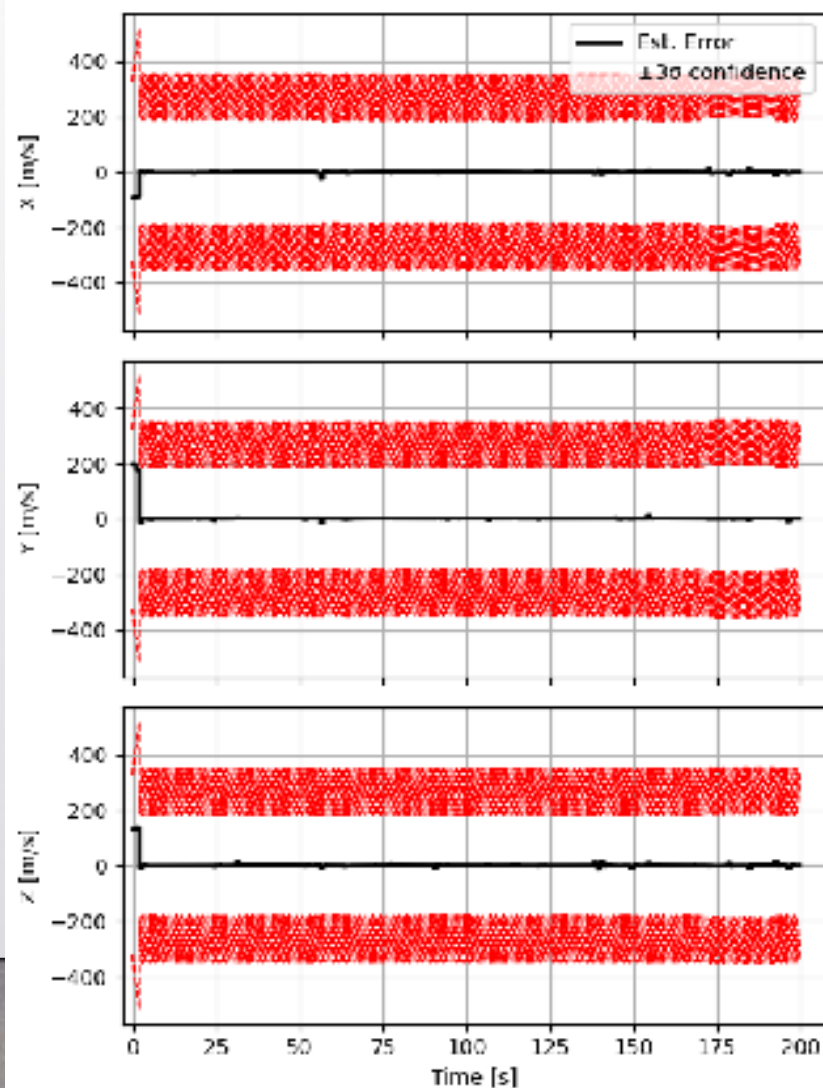
EGMF



EKF

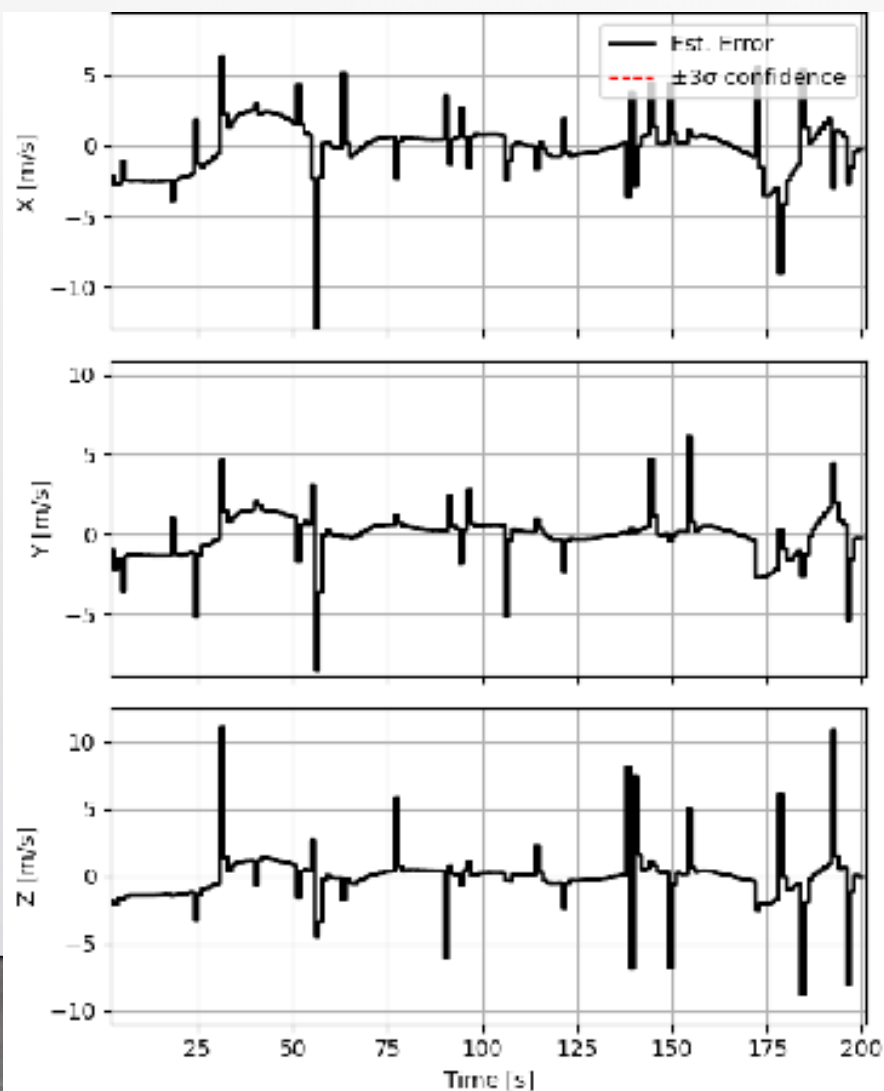


EGMF

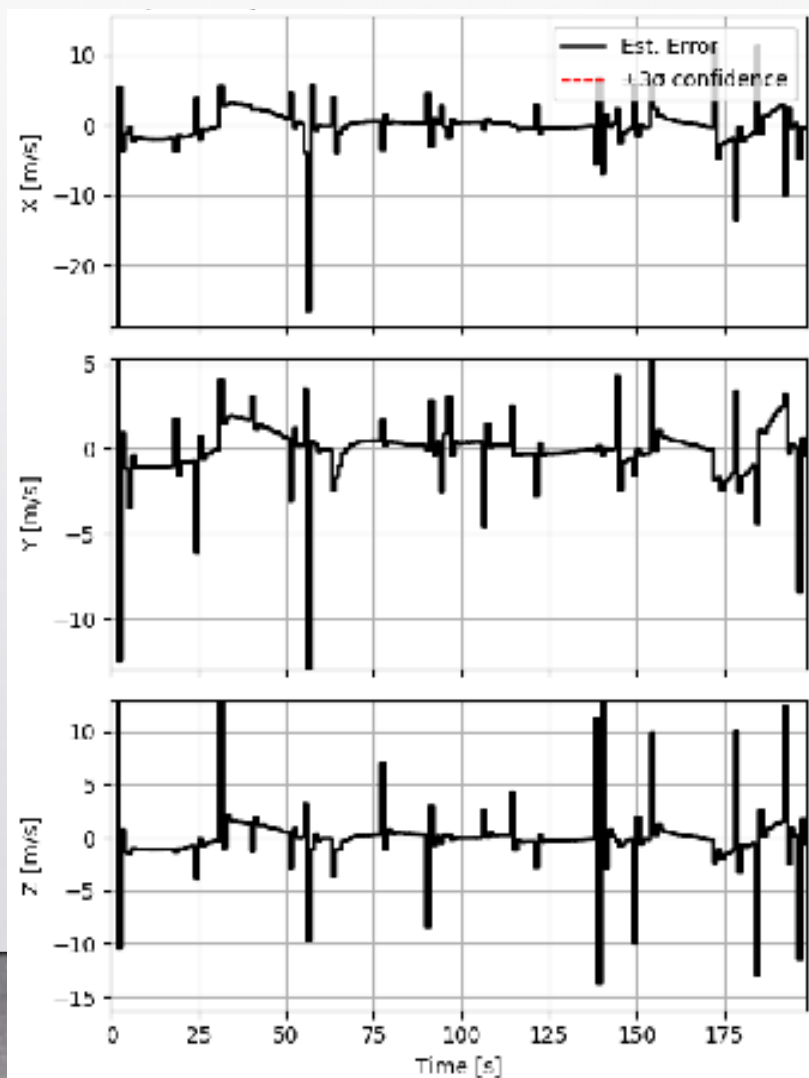


EGMF MCMF Frame Velocity State Error

EKF

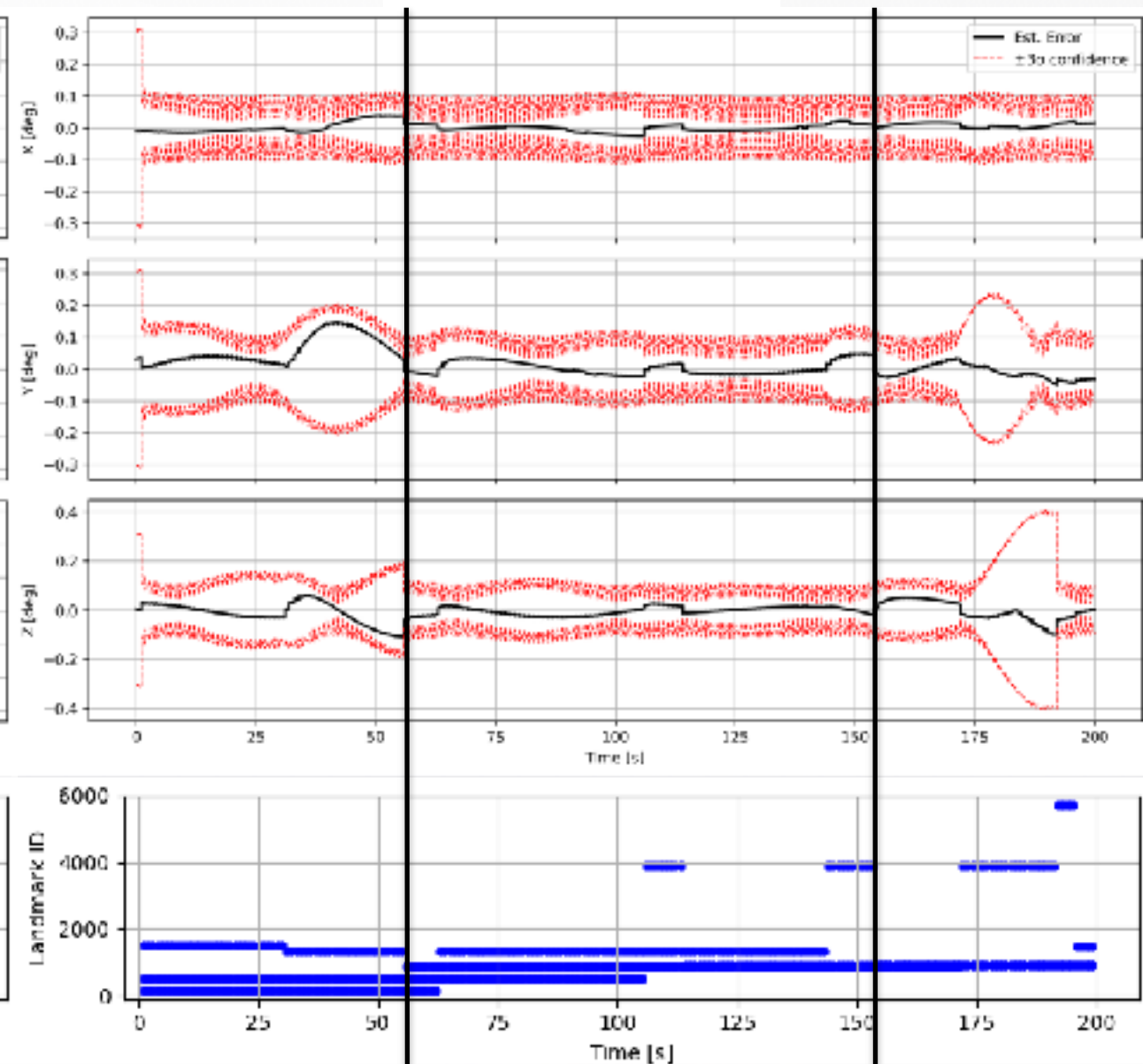
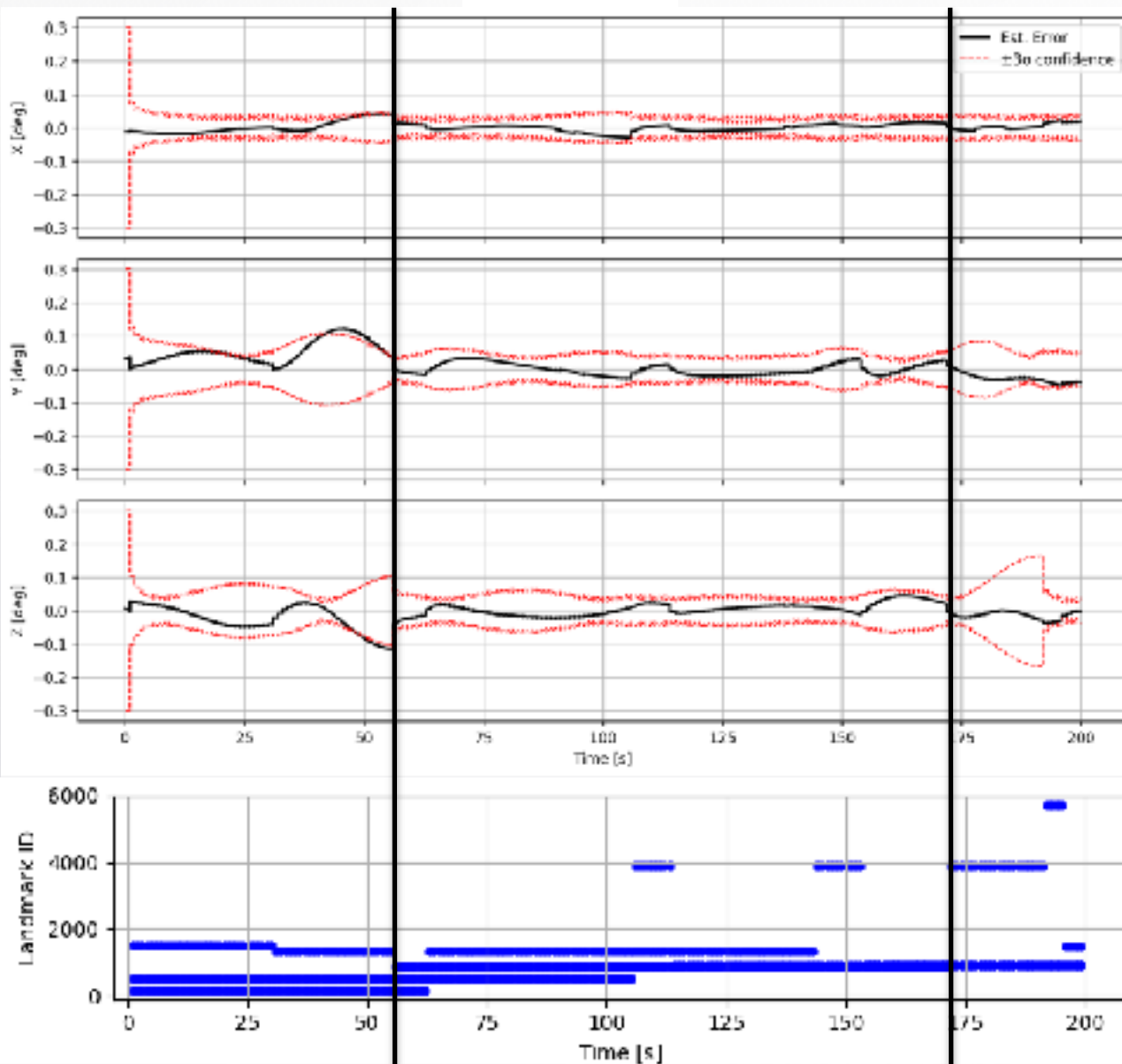


EGMF



Body Frame Attitude State Error

EKF



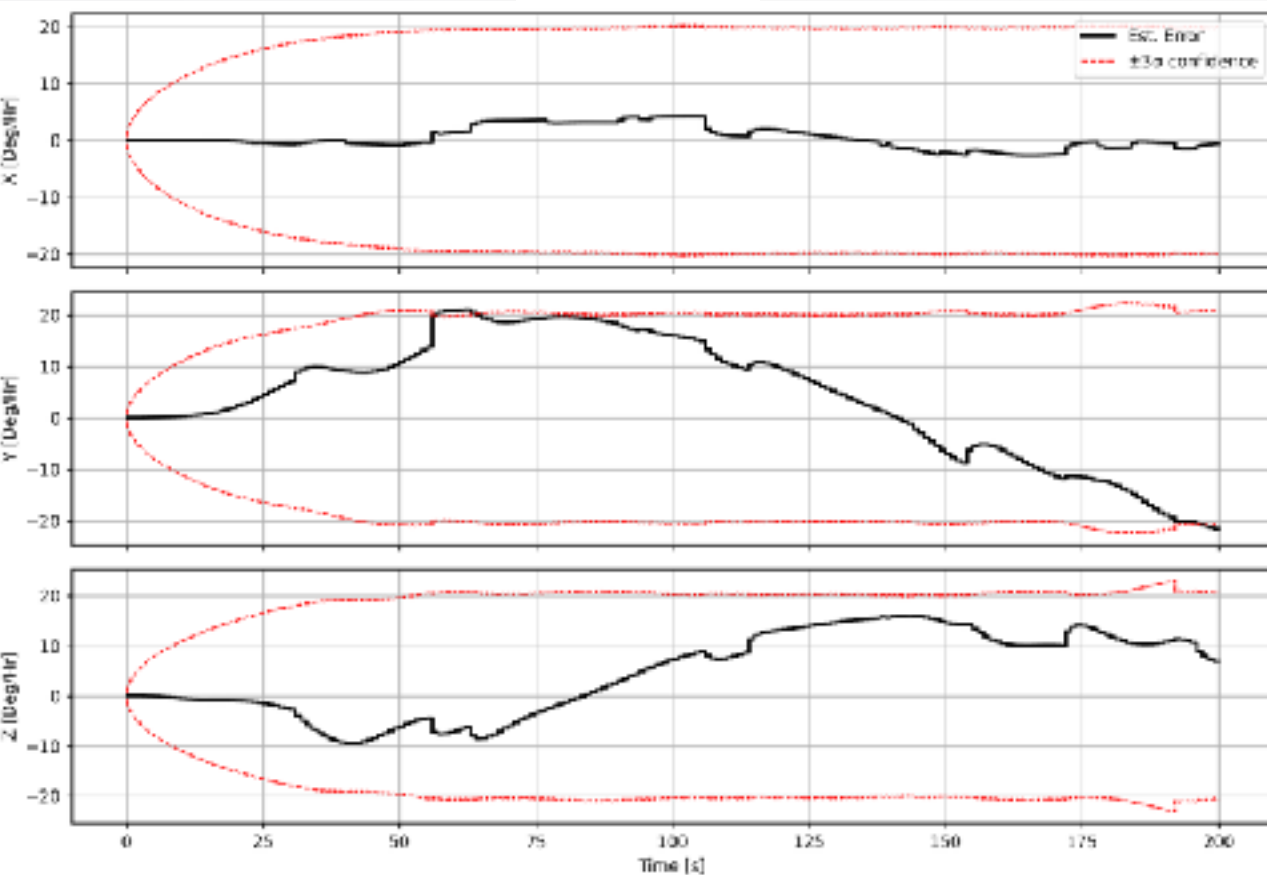
Gyroscope Bias State Error



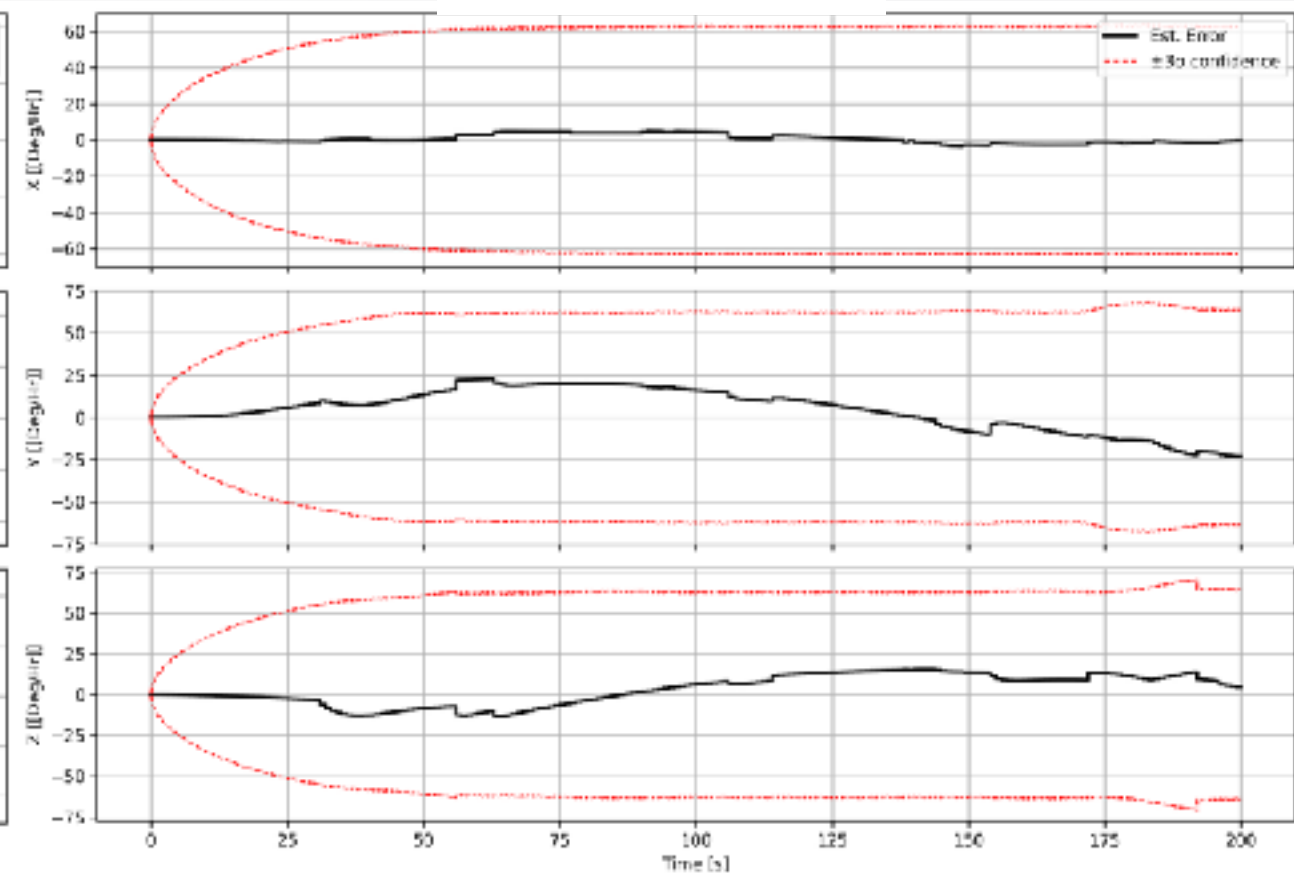
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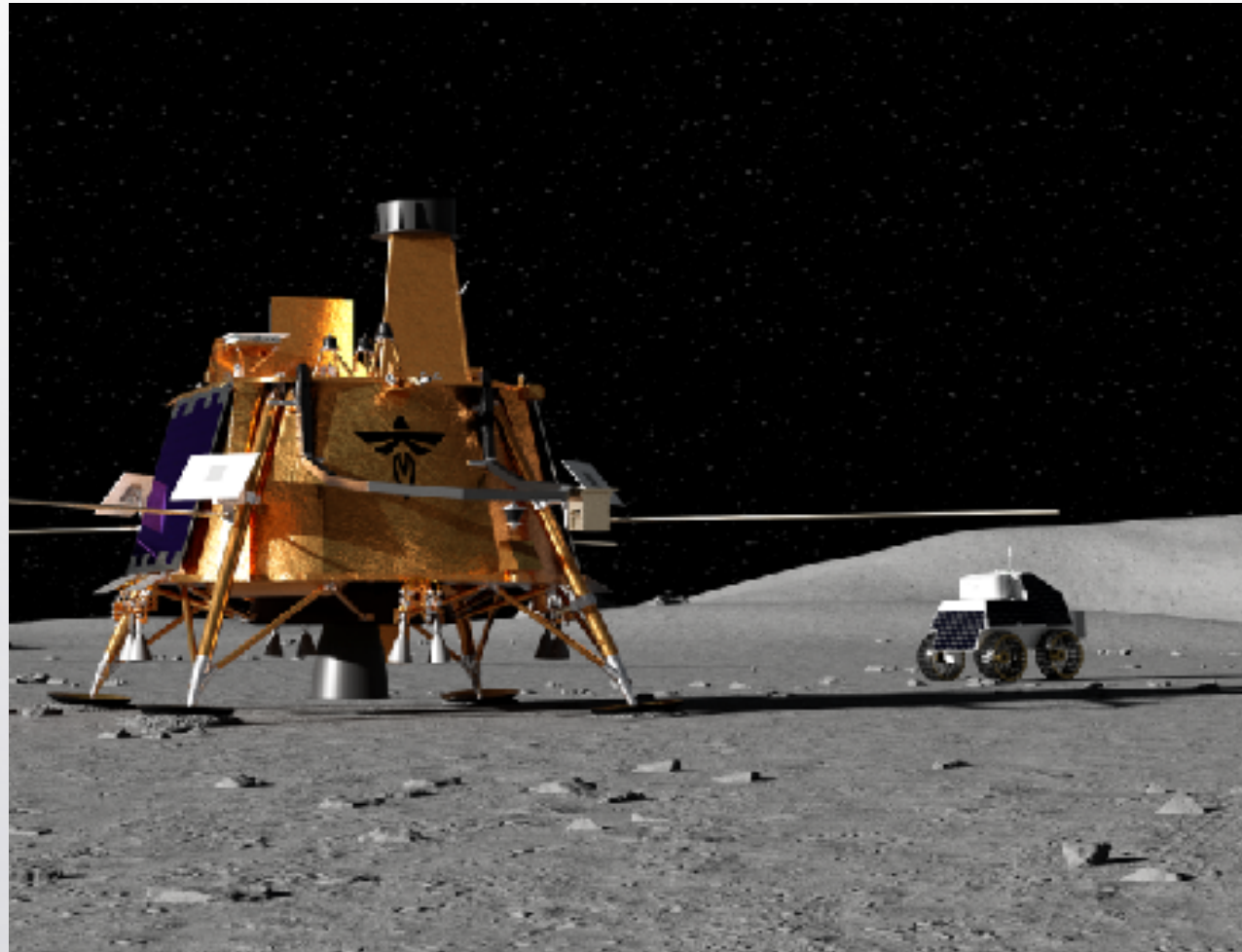
EKF



EGMF



Scenario 1: Posteriori State Error Statistics Comparison: EKF vs. EGMF



Scenario 2: MRP Attitude Control

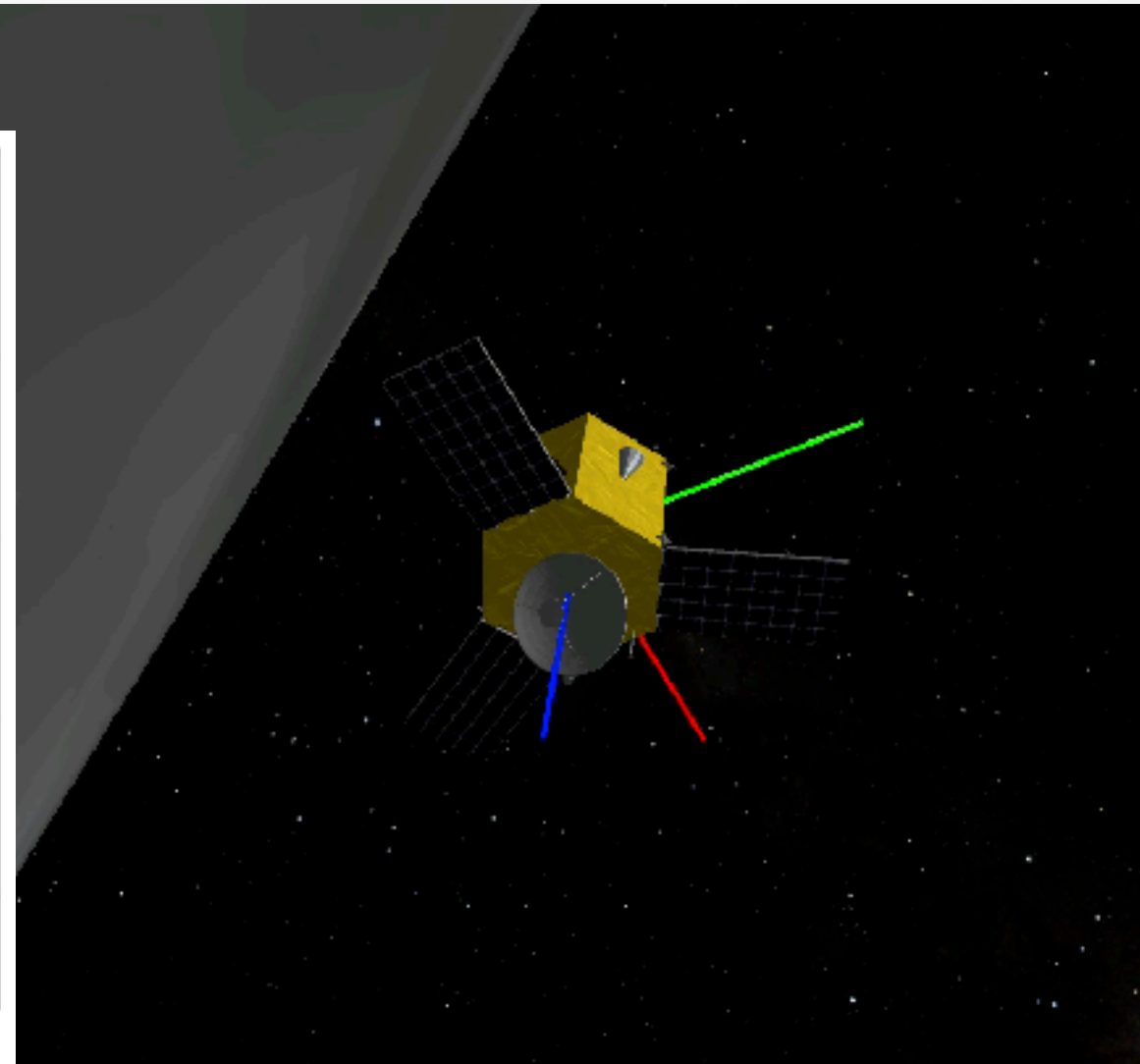
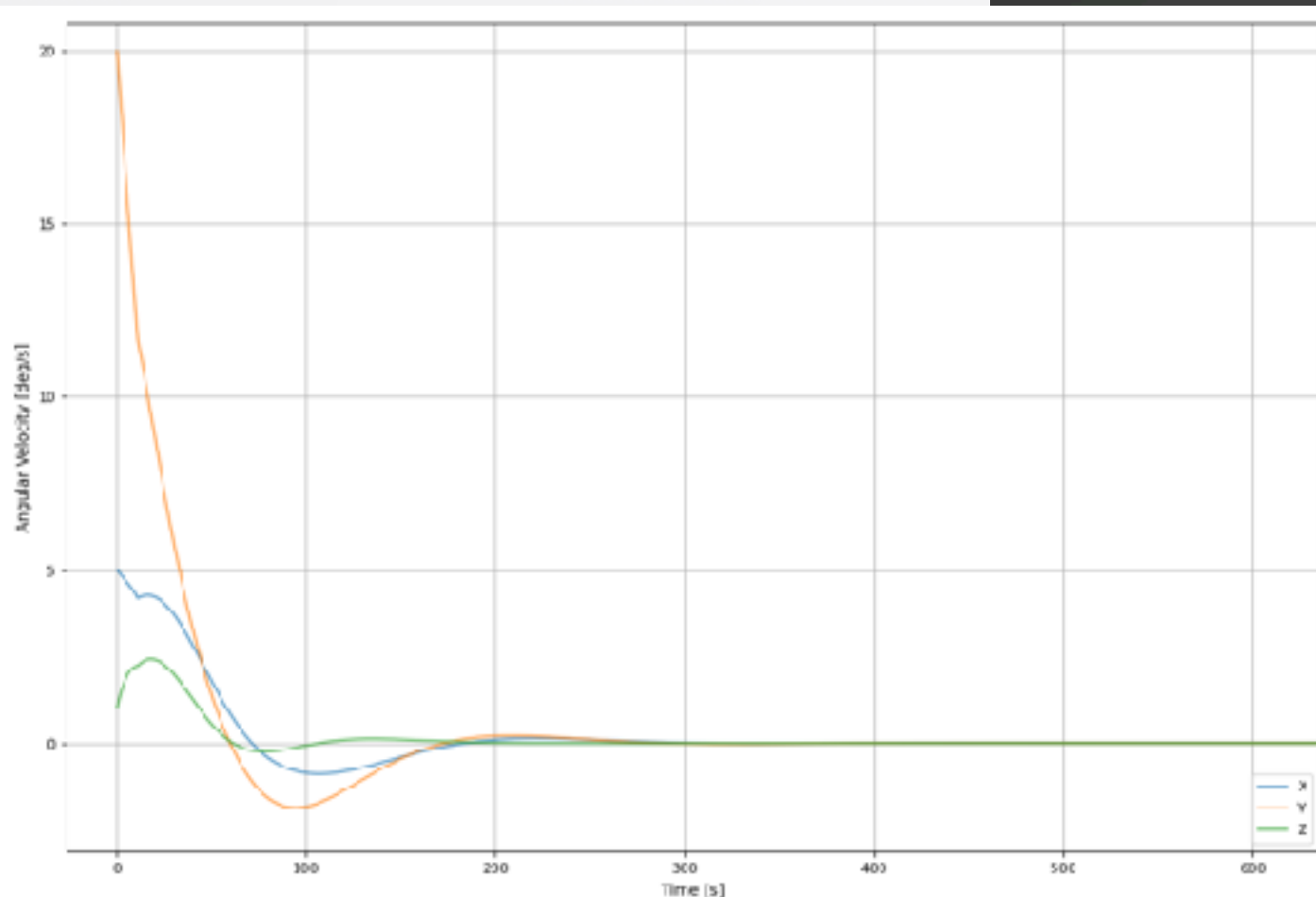


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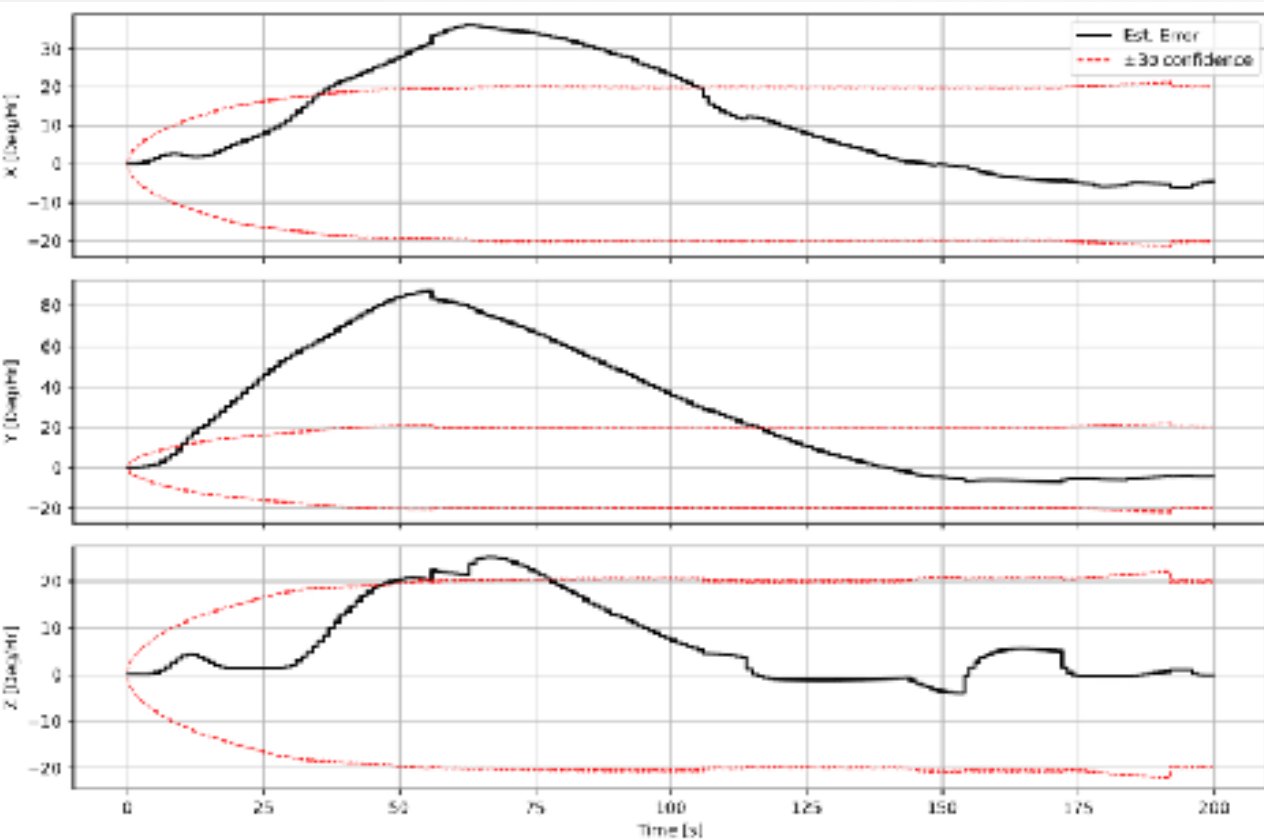
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$$\omega_{B/N}^B = [5.0, 20.0, 1.0]^T \text{ deg/s}$$

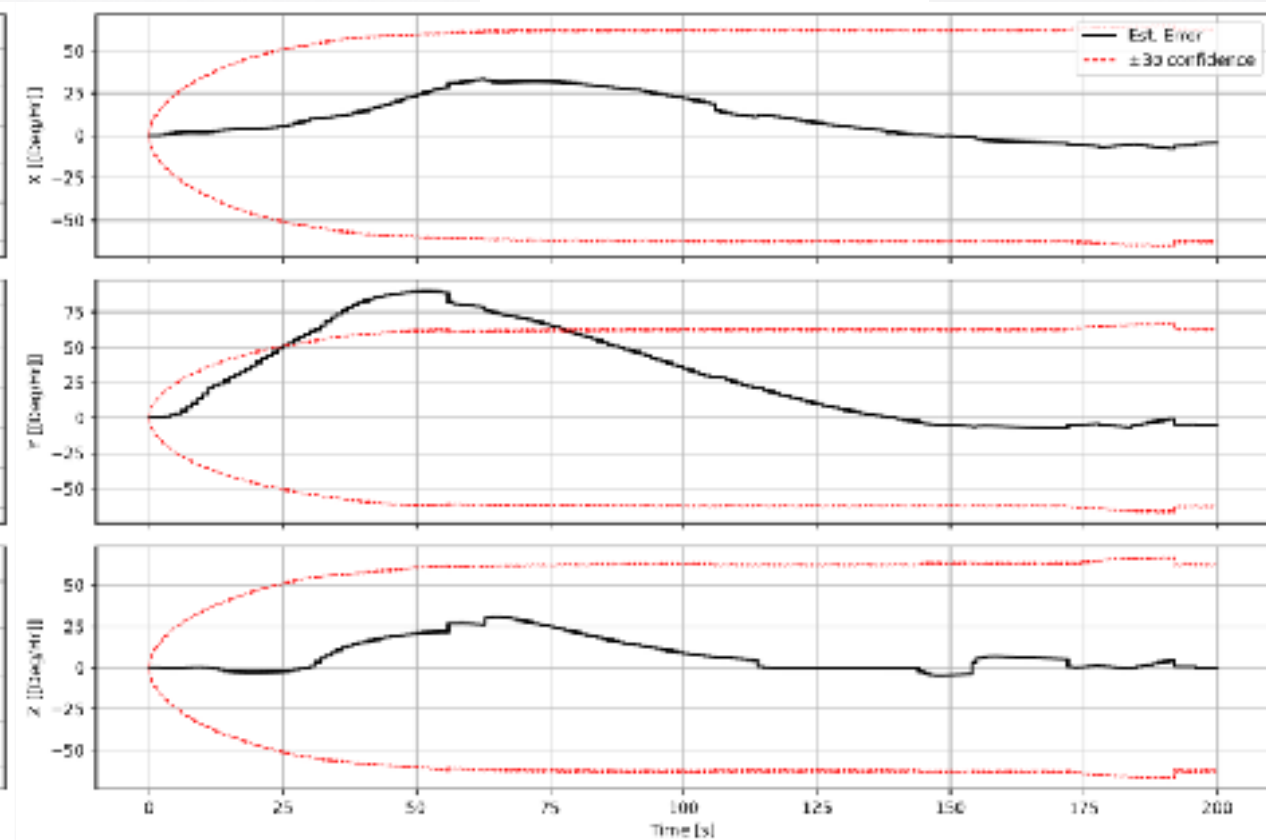
$$K_P = 3.5, K_d = 30.0.$$



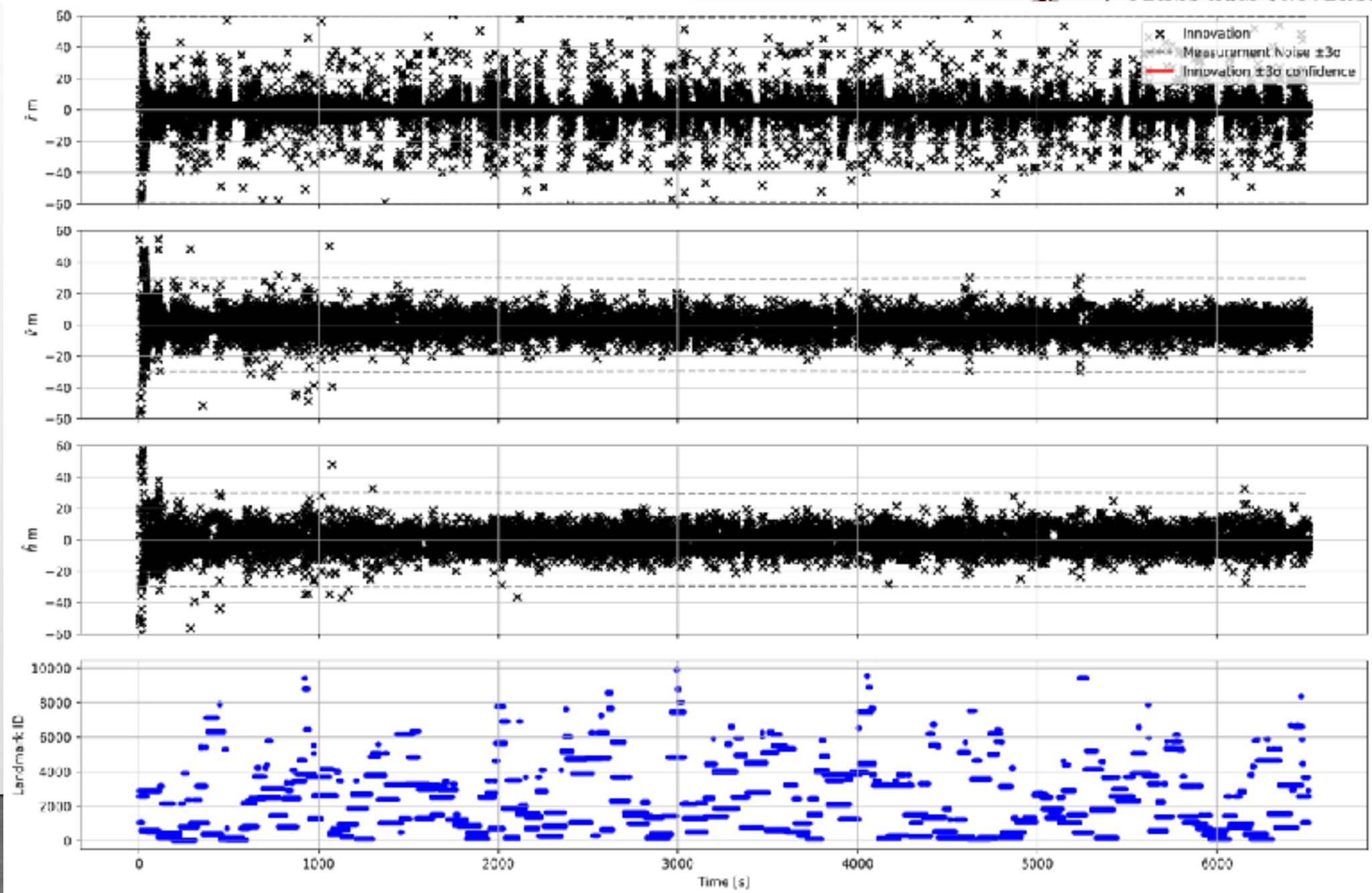
EKF



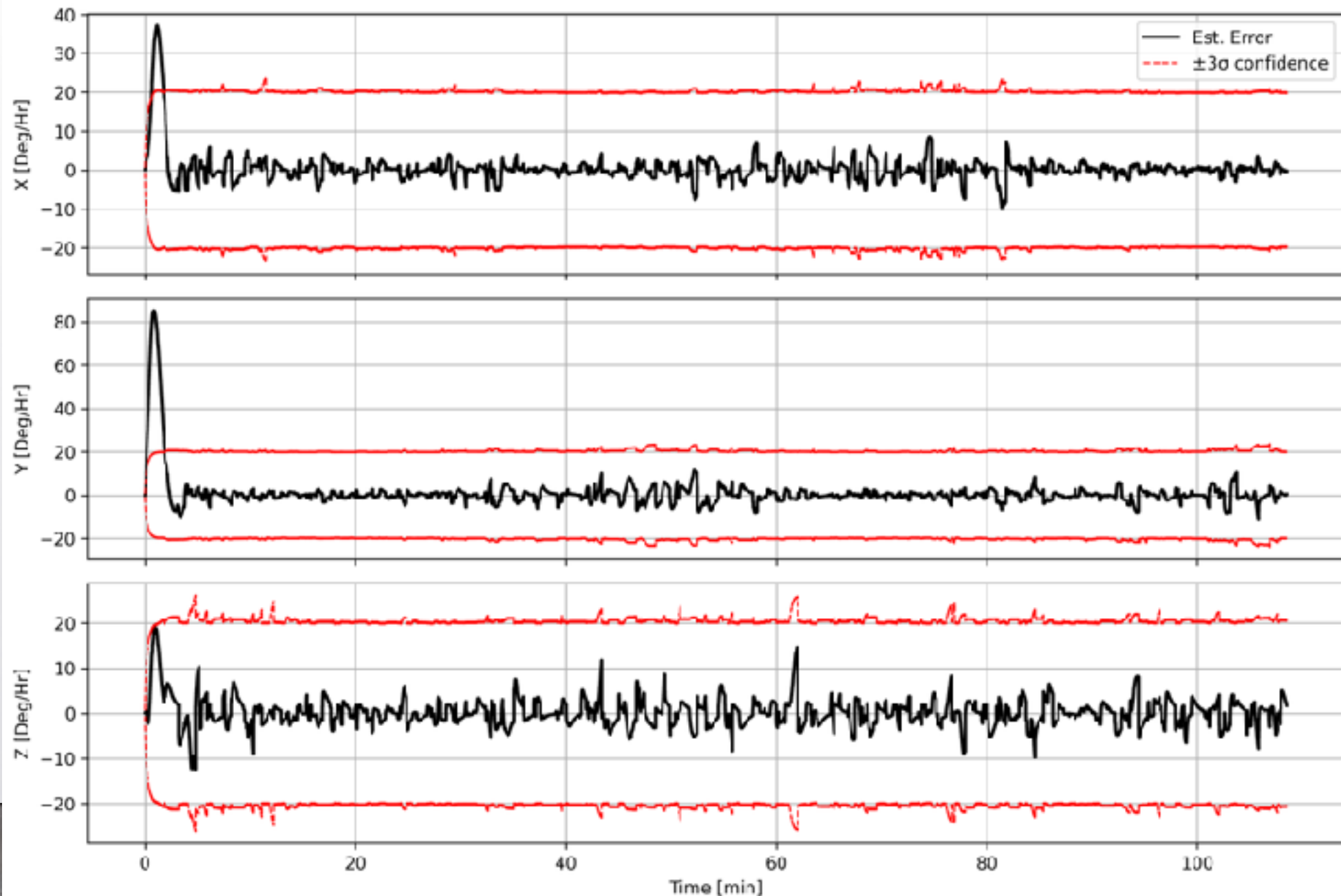
EGMF



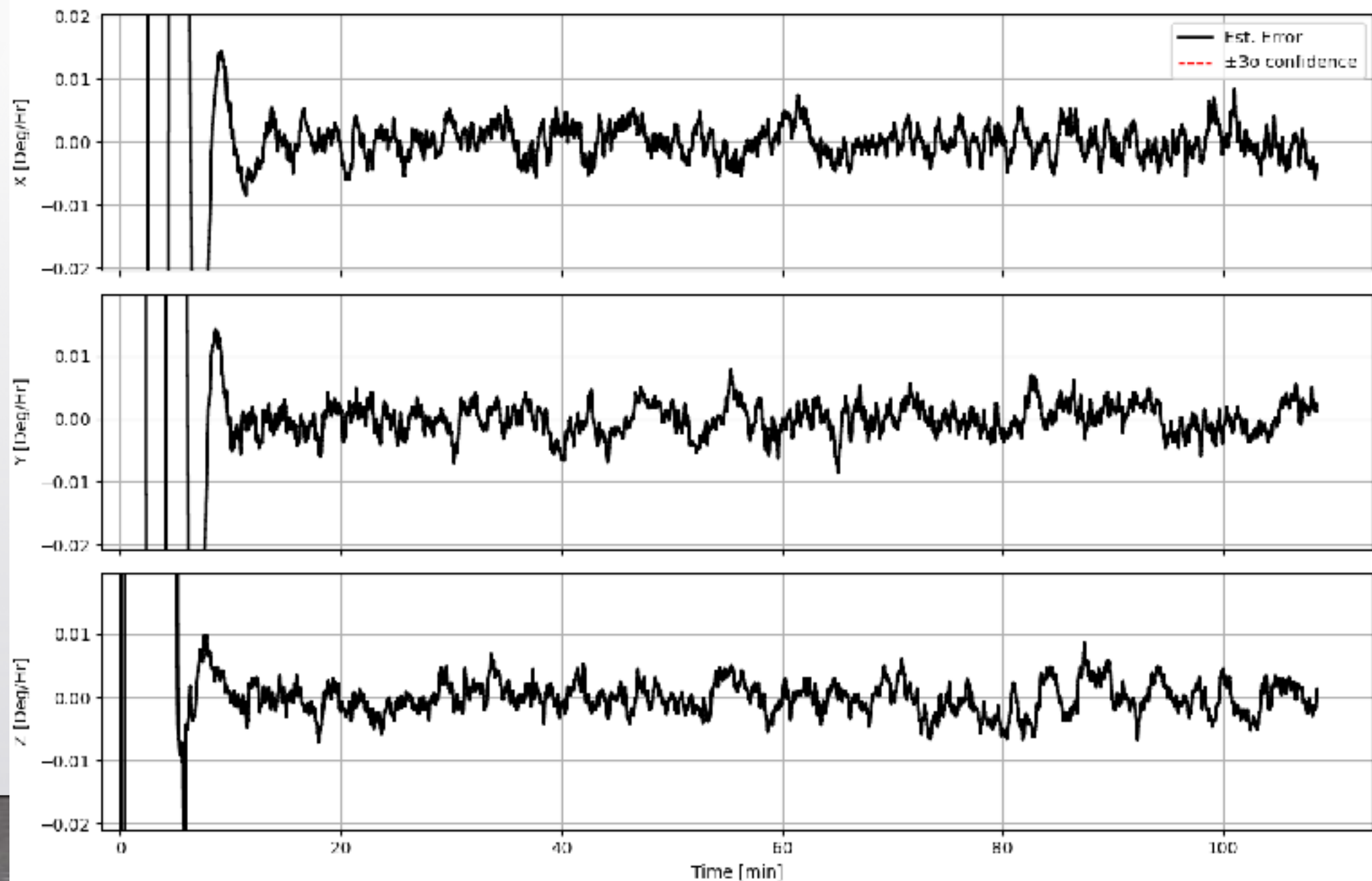
Scenario 2: One Orbital Period



Scenario 2: One Orbital Period EKF Gyroscope Bias State Error



Scenario 2: One Orbital Period True Measurements EKF Gyroscope Bias State Error



Conclusion

- EKF and EGMF obtained similar satisfactory lunar lander pose estimation results
- Relative position estimates continuously update translational states, but larger updates in multiplicative attitude states occur at epochs of landmark observation transition events
- Time and ability to estimate gyroscope bias is strongly correlated with measurement accuracy